

Assignment 1: MapReduce and Apache Spark

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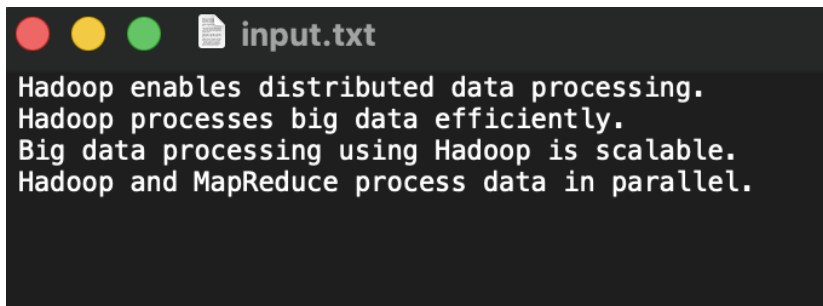
Course: Machine Learning With Big Data(CSL7110)

GitHub Repository and Pages Link: [Link](#)

Question 1 - Execution of WordCount Example

In this experiment, a single-node Hadoop cluster was tested by executing the standard WordCount MapReduce example. A sample text file was created locally and uploaded to HDFS, after which the WordCount program was executed using the Hadoop examples JAR file.

During execution, the mapper generated intermediate key–value pairs of the form (word, 1) for each word in the input text, and the reducer aggregated these values to compute the total frequency of each word. Upon successful completion, Hadoop created an output directory containing the _SUCCESS file and the part-r-00000 file with the final results, confirming the correct functioning of HDFS and the MapReduce processing framework.



```
input.txt
Hadoop enables distributed data processing.
Hadoop processes big data efficiently.
Big data processing using Hadoop is scalable.
Hadoop and MapReduce process data in parallel.
```

A custom input text file containing sample text with repeated words was created to demonstrate the working of the WordCount MapReduce program.

```
sushantakparasharjha@Sushantak ~ % hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount \
/user/$USER/input /user/$USER/output

2026-02-06 00:14:50,670 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-02-06 00:14:50,868 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-02-06 00:14:50,895 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-02-06 00:14:50,902 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-02-06 00:14:51,007 INFO input.FileInputFormat: Total input files to process : 1
2026-02-06 00:14:51,036 INFO mapreduce.JobSubmitter: number of splits:1
2026-02-06 00:14:51,095 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local687043458_0001
2026-02-06 00:14:51,095 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-06 00:14:51,146 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-02-06 00:14:51,147 INFO mapreduce.Job: Running job: job_local687043458_0001
2026-02-06 00:14:51,147 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-02-06 00:14:51,150 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-02-06 00:14:51,150 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-02-06 00:14:51,150 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-02-06 00:14:51,150 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-02-06 00:14:51,177 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-02-06 00:14:51,177 INFO mapred.LocalJobRunner: Starting task: attempt_local687043458_0001_m_000000_0
2026-02-06 00:14:51,180 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-02-06 00:14:51,184 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-02-06 00:14:51,184 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-02-06 00:14:51,188 INFO util.ProcfsBasedProcessTree: ProcfsBasedProcessTree currently is supported only on Linux.
2026-02-06 00:14:51,188 INFO mapred.Task: Using ResourceCalculatorProcessTree : null
2026-02-06 00:14:51,190 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/user/sushantakparasharjha/input/input.txt:0+177
2026-02-06 00:14:51,202 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-02-06 00:14:51,202 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-02-06 00:14:51,202 INFO mapred.MapTask: soft limit at 83860800
2026-02-06 00:14:51,202 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-02-06 00:14:51,241 INFO mapred.MapTask: bufstart = 0; bufend = 26214396; length = 6553600
2026-02-06 00:14:51,241 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-02-06 00:14:51,240 INFO mapred.LocalJobRunner:
2026-02-06 00:14:51,241 INFO mapred.MapTask: Starting flush of map output
2026-02-06 00:14:51,241 INFO mapred.MapTask: Spilling map output
2026-02-06 00:14:51,241 INFO mapred.MapTask: bufstart = 0; bufend = 272; bufvoid = 104857600
2026-02-06 00:14:51,241 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214304(104857216); length = 93/6553600
2026-02-06 00:14:51,253 INFO mapred.MapTask: Finished spill 0
2026-02-06 00:14:51,259 INFO mapred.Task: Task:attempt_local687043458_0001_m_000000_0 is done. And is in the process of committing
2026-02-06 00:14:51,260 INFO mapred.LocalJobRunner: map
2026-02-06 00:14:51,260 INFO mapred.Task: Task:attempt_local687043458_0001_m_000000_0 done.
2026-02-06 00:14:51,262 INFO mapred.Task: Final Counters for attempt_local687043458_0001_m_000000_0: Counters: 24
File System Counters
  FILE: Number of bytes read=281531
  FILE: Number of bytes written=942311
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=177
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=5
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=5
  Map output records=24
  Map output bytes=272
  Map output materialized bytes=254
  Input split bytes=120
  Combine input records=24
  Combine output records=18
```

Executing the WordCount MapReduce job

```
sushantakparasharjha@Sushantak ~ % hdfs dfs -ls /user/sushantakparasharjha/output

2026-02-06 00:15:50,076 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 sushantakparasharjha supergroup 0 2026-02-06 00:14 /user/sushantakparasharjha/output/_SUCCESS
-rw-r--r-- 1 sushantakparasharjha supergroup 176 2026-02-06 00:14 /user/sushantakparasharjha/output/part-r-00000
```

Listing the HDFS output directory after job completion

```
sushantakparasharjha@Sushantak ~ % hdfs dfs -cat /user/$USER/output/part-r-00000

2026-02-06 00:16:34,442 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Big 1
Hadoop 4
MapReduce 1
and 1
big 1
data 4
distributed 1
efficiently. 1
enables 1
in 1
is 1
parallel. 1
process 1
processes 1
processing 1
processing. 1
scalable. 1
using 1
sushantakparasharjha@Sushantak ~ % █
```

Displaying the reducer output (word frequencies)

Question 2: Map Phase Key–Value Pairs

In the Map phase, Hadoop reads each line of the input file and provides it to the mapper as a key–value pair where the key represents the byte offset of the line and the value represents the text content of the line. For the given lyrics example, the mapper receives input pairs such as:

```
(0, "We're up all night till the sun")
(31, "We're up all night to get some")
(63, "We're up all night for good fun")
(95, "We're up all night to get lucky")
```

The mapper processes each line, tokenizes it into words, and emits intermediate key–value pairs of the form (word, 1), for example:

```
("We're", 1), ("up", 1), ("all", 1), ("night", 1), ("till", 1), ("the", 1), ("sun", 1)
("We're", 1), ("up", 1), ("all", 1), ("night", 1), ("to", 1), ("get", 1), ("some", 1)
("We're", 1), ("up", 1), ("all", 1), ("night", 1), ("for", 1), ("good", 1), ("fun", 1)
("We're", 1), ("up", 1), ("all", 1), ("night", 1), ("to", 1), ("get", 1), ("lucky", 1)
```

Map Phase Types

Input Key Type: **LongWritable**

Input Value Type: **Text**

Output Key Type: **Text**

Output Value Type: **IntWritable**

Question 3: Reduce Phase Key–Value Pairs

After the Map phase, Hadoop performs the shuffle and sort operation, where intermediate key–value pairs having the same key are grouped together. The reducer receives grouped inputs such as:

```
("we're", [1, 1, 1, 1])
("up", [1, 1, 1, 1])
("all", [1, 1, 1, 1])
("night", [1, 1, 1, 1])
("to", [1, 1])
("get", [1, 1])
("till", [1])
```

```
("sun", [1])
("good", [1])
("fun", [1])
("lucky", [1])
```

The reducer sums the values associated with each word to generate the final word counts, producing outputs such as:

```
("we're", 4), ("up", 4), ("all", 4), ("night", 4), ("to", 2), ("get", 2), ("till", 1), ("sun", 1), ("good", 1),
("fun", 1), ("lucky", 1)
```

Reduce Phase Types

Input Key Type: **Text**

Input Value Type: **Iterable<IntWritable>**

Output Key Type: **Text**

Output Value Type: **IntWritable**

Question 4 — Filling the Placeholders in WordCount.java

From the Hadoop MapReduce tutorial WordCount program, the placeholders in the Map and Reduce function definitions are replaced using the data types identified in Question 2.

Map function definition

Java

```
public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException {
    ...
}
```

Reduce function definition

Java

```
public void reduce(Text key, Iterable<IntWritable> values, Context context)
    throws IOException, InterruptedException {
    ...
}
```

Job output key and value classes

Java

```
job.setOutputKeyClass(Text.class);  
job.setOutputValueClass(IntWritable.class);
```

Explanation:

The Mapper receives input key–value pairs of type (LongWritable, Text) and emits intermediate (Text, IntWritable) pairs. The Reducer processes grouped values of type Iterable<IntWritable> and produces the final output of type (Text, IntWritable).

Question 5 — Writing the map() Function

The map() function removes punctuation using String.replaceAll(), splits the input line into words using StringTokenizer, and emits (word, 1) pairs.

Java

```
public void map(LongWritable key, Text value, Context context)  
    throws IOException, InterruptedException {  
  
    String line = value.toString().replaceAll("[^a-zA-Z]", " ");  
    StringTokenizer itr = new StringTokenizer(line);  
  
    while (itr.hasMoreTokens()) {  
        Text word = new Text(itr.nextToken().toLowerCase());  
        context.write(word, new IntWritable(1));  
    }  
}
```

Explanation:

The map() function processes each input line, removes punctuation characters, tokenizes the cleaned line into words, and emits intermediate key–value pairs (word, 1) for each word. This prepares the data for aggregation in the Reduce phase.

Question 6 — Writing the reduce() Function

The reduce() function receives each word along with a collection of integer values generated by the Mapper. It sums these values to compute the total number of occurrences of each word and outputs the final result.

reduce() Function

Java

```
public void reduce(Text key, Iterable<IntWritable> values,
                  Context context)
    throws IOException, InterruptedException {

    int sum = 0;

    // Sum all values associated with the word
    for (IntWritable val : values) {
        sum += val.get();
    }

    // Emit final word count
    context.write(key, new IntWritable(sum));
}
```

Explanation:

The `reduce()` function processes grouped intermediate key–value pairs produced during the Map phase. For each word (key), it iterates through the list of associated integer values (`Iterable<IntWritable>`), calculates their total, and outputs the final word frequency as (word, total_count).

Question 7 — Compile and Run the WordCount Program

After implementing the Mapper and Reducer functions, the `WordCount.java` program was compiled using the Hadoop classpath and packaged into a JAR file. The dataset `200.txt` was uploaded to the Hadoop Distributed File System (HDFS), and the custom WordCount MapReduce job was executed. Since Hadoop does not allow overwriting an existing output directory, the previous output folder was removed before running the job again. After successful execution, the reducer output files were merged into a single file (`result.txt`), which contains the final word frequency counts generated by the program.

```
sushantakparasharjha@Sushantak wordcount_project % javac -classpath `hadoop classpath` WordCount.java
```

Compilation of the `WordCount.java` program using the Hadoop classpath.


```
sushantakparasharjha@Sushantak wordcount_project % jar cf WordCount.jar WordCount*.class
```

Creation of the executable WordCount.jar file containing the compiled classes.

```
sushantakparasharjha@Sushantak wordcount_project % hadoop fs -copyFromLocal 200.txt
```

```
2026-02-06 12:19:38,979 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...  
copyFromLocal: `200.txt': File exists
```

Copying the dataset file 200.txt from the local filesystem to the Hadoop Distributed File System (HDFS) using the hadoop fs -copyFromLocal command.

```
sushantakparasharjha@Sushantak wordcount_project % hadoop jar WordCount.jar WordCount 200.txt output
```

```
2026-02-06 12:04:59,109 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java  
2026-02-06 12:04:59,310 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties  
2026-02-06 12:04:59,339 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).  
2026-02-06 12:04:59,339 INFO impl.MetricsSystemImpl: JobTracker metrics system started  
2026-02-06 12:04:59,389 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool  
2026-02-06 12:04:59,438 INFO input.FileInputFormat: Total input files to process : 1  
2026-02-06 12:04:59,481 INFO mapreduce.JobSubmitter: number of splits:1  
2026-02-06 12:04:59,538 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local431806245_0001  
2026-02-06 12:04:59,538 INFO mapreduce.JobSubmitter: Executing with tokens: []  
2026-02-06 12:04:59,585 INFO mapreduce.Job: The url to track the job: http://localhost:8080/  
2026-02-06 12:04:59,585 INFO mapreduce.Job: Running job: job_local431806245_0001  
2026-02-06 12:04:59,586 INFO mapred.LocalJobRunner: OutputCommitter set in config null  
2026-02-06 12:04:59,588 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCom  
2026-02-06 12:04:59,588 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2  
2026-02-06 12:04:59,588 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directo  
2026-02-06 12:04:59,588 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter  
2026-02-06 12:04:59,613 INFO mapred.LocalJobRunner: Waiting for map tasks  
2026-02-06 12:04:59,613 INFO mapred.LocalJobRunner: Starting task: attempt_local431806245_0001_m_000000_0  
2026-02-06 12:04:59,620 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCom  
2026-02-06 12:04:59,620 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2  
2026-02-06 12:04:59,620 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directo  
2026-02-06 12:04:59,623 INFO util.ProcfsBasedProcessTree: ProcfsBasedProcessTree currently is supported only on Linux.  
2026-02-06 12:04:59,624 INFO mapred.Task: Using ResourceCalculatorProcessTree : null  
2026-02-06 12:04:59,625 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/user/sushantakparasharjha/200.txt:0+8312639  
2026-02-06 12:04:59,643 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)  
2026-02-06 12:04:59,643 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100  
2026-02-06 12:04:59,643 INFO mapred.MapTask: soft limit at 83886080  
2026-02-06 12:04:59,643 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600  
2026-02-06 12:04:59,643 INFO mapred.MapTask: kvstart = 26214396; length = 6553600  
2026-02-06 12:04:59,644 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer  
2026-02-06 12:05:00,031 INFO mapred.LocalJobRunner:  
2026-02-06 12:05:00,032 INFO mapred.MapTask: Starting flush of map output  
2026-02-06 12:05:00,032 INFO mapred.MapTask: Spilling map output  
2026-02-06 12:05:00,032 INFO mapred.MapTask: bufstart = 0; bufend = 12867576; bufvoid = 104857600  
2026-02-06 12:05:00,032 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 20939320(83757280); length = 5275077/6553600  
2026-02-06 12:05:00,404 INFO mapred.MapTask: Finished spill 0  
2026-02-06 12:05:00,412 INFO mapred.Task: Task:attempt_local431806245_0001_m_000000_0 is done. And is in the process of committin  
2026-02-06 12:05:00,413 INFO mapred.LocalJobRunner: map  
2026-02-06 12:05:00,413 INFO mapred.Task: Task 'attempt_local431806245_0001_m_000000_0' done.
```

Execution of the custom WordCount MapReduce job on the dataset stored in HDFS.

```
sushantakparasharjha@Sushantak wordcount_project % hdfs dfs -getmerge output result.txt
cat result.txt

2026-02-06 12:07:18,223 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
a      26012
aa     10
aaa    1
aabborgt      1
aachen 7
aachens 1
aad      1
aadms    1
aagesen 2
aahmes    2
aak       1
aal       1
aalborg 4
aalen    2
aalesund    4
aali      2
aalst     1
aar      13
aarau    12
aarberg  1
aarburg  2
aard      6
aardrjko   1
aare      1
aargau   10
aargauer   1
aargaus  1
aarhus    5
aaron    19
aaronite   1
aaronites  1
aarssen  1
aarssens  2
aartsen  1
```

Reducer output merged into a single file showing the final word frequency counts.

Question 8 - Copying data to HDFS

```
sushantakparasharjha@Sushantak wordcount_project % hadoop fs -copyFromLocal 200.txt

2026-02-06 12:19:38,979 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
copyFromLocal: '200.txt': File exists
```

Copying the dataset file 200.txt from the local filesystem to the Hadoop Distributed File System (HDFS) using the `hadoop fs -copyFromLocal` command.

```
sushantakparasharjha@Sushantak wordcount_project % hadoop fs -ls

2026-02-06 12:22:02,937 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
using builtin-java classes where applicable
Found 3 items
-rw-r--r--  1 sushantakparasharjha supergroup  8312639 2026-02-06 12:02 200.txt
drwxr-xr-x  - sushantakparasharjha supergroup      0 2026-02-06 00:14 input
drwxr-xr-x  - sushantakparasharjha supergroup      0 2026-02-06 12:05 output
```


Listing the contents of the HDFS directory using the `hadoop fs -ls` command, showing the uploaded file along with the existing directories and their replication information.

The `hadoop fs -ls` command lists the files and directories stored in the Hadoop Distributed File System (HDFS). In the displayed output, the first entry represents the file `200.txt`, while the other entries (input and output) represent directories.

The second column in the listing shows the replication factor for files. For `200.txt`, the replication factor is 1, meaning that one copy of the file is stored in HDFS.

Why don't we have a replication factor for directories?

-> Directories do not display a replication factor because they do not store actual data blocks; instead, they contain metadata describing the files stored inside them.

The replication factor determines the reliability and performance of the system. Increasing the replication factor improves fault tolerance and read performance because multiple copies of the file are available across different nodes, but it increases storage consumption and write overhead. Decreasing the replication factor reduces storage usage but lowers fault tolerance in case of node failures.

Question 9 - Total execution time of the job

To measure the execution time of the WordCount MapReduce program, timing statements were added in the `main()` method using `System.currentTimeMillis()`. The start time was recorded before submitting the job and the end time after job completion. The difference between these values was printed to the console, representing the total execution time of the MapReduce job.

```
sushantakparasharjha@Sushantak wordcount_project % hadoop jar WordCount.jar WordCount 200.txt output
2026-02-06 16:08:23,108 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
using builtin-java classes where applicable
2026-02-06 16:08:23,331 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-02-06 16:08:23,362 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-02-06 16:08:23,363 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-02-06 16:08:23,423 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed.
Implement the Tool interface and execute your application with ToolRunner to remedy this.
```

```
File Input Format Counters
      Bytes Read=8312639
File Output Format Counters
      Bytes Written=675870
Execution time: 2360 ms
```

Executing the WordCount MapReduce job with execution time measurement enabled.

```
sushantakparasharjha@Sushantak wordcount_project % hadoop jar WordCount.jar WordCount 200.txt output

2026-02-06 16:20:49,826 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
using builtin-java classes where applicable
2026-02-06 16:20:50,051 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-02-06 16:20:50,084 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-02-06 16:20:50,084 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-02-06 16:20:50,142 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed.
Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-02-06 16:20:50,190 INFO input.FileInputFormat: Total input files to process : 1
```

```
File Input Format Counters
  Bytes Read=8312639
File Output Format Counters
  Bytes Written=675870
Execution time: 2338 ms
```

Execution time of the WordCount MapReduce job after modifying the input split size configuration parameter.

How does changing the value of that parameter impact performance? Why?

-> The parameter `mapreduce.input.fileinputformat.split.maxsize` determines the maximum size of each input split, which directly affects the number of mapper tasks created during job execution. Increasing the split size reduces the number of mapper tasks, which decreases task scheduling overhead but also reduces parallelism, potentially slowing down processing in distributed environments. Decreasing the split size increases the number of mapper tasks, improving parallelism and allowing more data to be processed simultaneously, but excessive splitting can increase overhead due to the creation and management of many tasks. Therefore, the performance of a MapReduce job depends on selecting an optimal split size that balances parallelism and task management overhead.

Question 10 - Book Metadata Extraction and Analysis

The collection of Project Gutenberg books was loaded into Apache Spark using the text file reader. Each line of every book was stored as a row in a DataFrame along with the corresponding file name. Regular expressions were then used to extract metadata fields such as title, release date, language, and encoding from the book headers.

The extracted metadata was organized into a structured DataFrame to enable further analysis. Using this metadata, additional exploratory computations were performed, including the number of books released per year, the most common language in the dataset, and the average title length. This preprocessing step prepared the dataset for subsequent text processing and similarity analysis tasks.

```
>>> from pyspark.sql.functions import *
...
... # Load books dataset
... books_df = spark.read.text("file:///Users/sushantakparasharjha/D184MB/") \
...     .withColumnRenamed("value","text") \
...     .withColumn("file_name", input_file_name())
...
... books_df.show(5, False)
...
```

text	file_name
The Project Gutenberg EBook of The Project Gutenberg Gutenberg Encyclopedia, Vol 1, by Project Gutenberg	file:///Users/sushantakparasharjha/D184MB/200.txt
This eBook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever. You may copy it, give it away or	file:///Users/sushantakparasharjha/D184MB/200.txt

only showing top 5 rows

Loading the Project Gutenberg books dataset into a Spark DataFrame and displaying sample records

```
... # Metadata extraction
... metadata_df = books_df \
...     .filter(
...         lower(col("text")).startswith("title:") |
...         lower(col("text")).startswith("release date:") |
...         lower(col("text")).startswith("language:") |
...         lower(col("text")).startswith("encoding:")
...     ) \
...     .groupBy("file_name") \
...     .agg(
...         max(when(lower(col("text")).startswith("title:"), col("text"))).alias("title"),
...         max(when(lower(col("text")).startswith("release date:"), col("text"))).alias("release_date"),
...         max(when(lower(col("text")).startswith("language:"), col("text"))).alias("language"),
...         max(when(lower(col("text")).startswith("encoding:"), col("text"))).alias("encoding")
...     )
...
... metadata_df.show(5, False)
...
```

file_name	title	release_date	language	encoding
file:///Users/sushantakparasharjha/D184MB/10.txt	Title: The King James Bible	[Release Date: March 2, 2011 [EBook #10]	Language: English	NULL
file:///Users/sushantakparasharjha/D184MB/101.txt	Title: Hacker Crackdown	[Release Date: January, 1994	Language: English	NULL
file:///Users/sushantakparasharjha/D184MB/102.txt	Title: The Tragedy of Pudd'nhead Wilson	[Release Date: January, 1994	Language: English	NULL
file:///Users/sushantakparasharjha/D184MB/103.txt	Title: Around the World in 80 Days	[Release Date: May 15, 2008 [EBook #103]	Language: English	NULL
file:///Users/sushantakparasharjha/D184MB/104.txt	Title: Franklin Delano Roosevelt's First Inaugural Address	[Release Date: May 14, 2008 [EBook #104]	Language: English	NULL

only showing top 5 rows

Extracting metadata fields (title, release date, language, encoding) from the book headers using regular expressions and displaying the resulting metadata_df.

```
scala> val books_per_year = metadata_df
      |   .withColumn("year", regexp_extract(col("release_date"), "(\\d{4})", 1))
      |   .filter(col("year") != "")
      |   .groupBy("year")
      |   .count()
      |   .orderBy("year")
      |   books_per_year.show()
+-----+-----+
|year|count|
+-----+-----+
|1975|    1|
|1978|    1|
|1979|    1|
|1991|    7|
|1992|   19|
|1993|   13|
|1994|   17|
|1995|   60|
|1996|   53|
|2002|    1|
|2004|    7|
|2005|    4|
|2006|   42|
|2007|   13|
|2008|  154|
|2009|    1|
|2010|    9|
|2011|    1|
|2012|    2|
|2013|    1|
+-----+-----+
only showing top 20 rows
val books_per_year: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [year: string, count: bigint]
```

Calculating the number of books released each year.

```
... # Books released each year
... books_per_year = metadata_df \
...   .withColumn("year", regexp_extract(col("release_date"), "(\\d{4})", 1)) \
...   .groupBy("year") \
...   .count()
...
... books_per_year.show()
```

only showing top 20 rows

```
+-----+-----+
|year|count|
+-----+-----+
|2012|    2|
|2013|    1|
|2005|    4|
|NULL|    2|
|1978|    1|
|2002|    1|
|2009|    1|
|1995|   60|
|2006|   42|
|2004|    7|
|2011|    1|
|1992|   19|
|2008|  154|
|1994|   17|
|2007|   13|
|1996|   53|
|1979|    1|
|2015|    1|
|1993|   13|
|    |    1|
+-----+-----+
```

only showing top 20 rows


```

... # Most common language
... metadata_df.groupBy("language") \
...     .count() \
...     .orderBy(desc("count")) \
...     .show(5)

```

language	count
Language: English	403
Language: Latin	6
Language: total p...	1
language: and thu...	1

Finding the most common language in the dataset.

```

... # Average title length
... metadata_df \
...     .withColumn("title_length", length(col("title"))) \
...     .select(avg("title_length")) \
...     .show()

```

avg(title_length)
29.829268292682926

Calculating the average length of book titles (in characters).

Explain the regular expressions you used to extract the title, release date, language, and encoding. Discuss any challenges or limitations in using regular expressions for this task.

-> Regular expressions were used to identify metadata lines in the text that begin with specific keywords such as "Title:", "Release Date:", "Language:", and "Encoding:". Pattern matching was performed using expressions that check whether a line starts with these labels, for example `^Title:` or `^Release Date:`. For extracting the year from the release date, the regular expression `(\d{4})` was used to capture the four-digit year present in the release date string.

One challenge in using regular expressions for metadata extraction is that the formatting of Project Gutenberg files is not always consistent. Some books may use slightly different label formats, contain additional spaces, or place metadata in different sections of the text. As a result, certain metadata values may not be captured correctly or may be missing. Additionally, some files may not contain all metadata fields, leading to null or incomplete records in the resulting DataFrame.

What are some potential issues with the extracted metadata (e.g., inconsistencies, missing values)? How would you handle these issues in a real-world scenario?

-> In real-world scenarios, these issues can be handled by applying additional data-cleaning steps such as normalizing text formatting, using multiple pattern variations to capture different label styles, filling missing values with default placeholders, and performing validation checks to ensure extracted metadata is consistent and complete before further analysis.

Question 11 - TF-IDF and Book Similarity

```
>>> from pyspark.sql.functions import input_file_name, regexp_replace, lower, col, desc
... from pyspark.ml.feature import Tokenizer, StopWordsRemover, CountVectorizer, IDF, Normalizer
... from pyspark.sql.types import DoubleType
... from pyspark.sql.functions import udf
...
... # Load books
... books_df = spark.read.text("file:///Users/sushantakparasharjha/D184MB/") \
...     .withColumnRenamed("value", "text") \
...     .withColumn("file_name", input_file_name())
...
... books_df.show(5, False)
...
... # Clean text
... clean_df = books_df.withColumn(
...     "clean_text",
...     lower(regexp_replace(col("text"), "[^a-zA-Z\\s]", " "))
... )
...
... clean_df.select("clean_text").show(5, False)
...
... 
```

text	file_name
The Project Gutenberg EBook of The Project Gutenberg Gutenberg Encyclopedia, Vol 1, by Project Gutenberg	file:///Users/sushantakparasharjha/D184MB/200.txt
This eBook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever. You may copy it, give it away or	file:///Users/sushantakparasharjha/D184MB/200.txt
only showing top 5 rows	
clean_text	
the project gutenberg ebook of the project gutenberg gutenberg encyclopedia vol by project gutenberg	
this ebook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever you may copy it give it away or	
only showing top 5 rows	

Cleaning the book text by converting to lowercase and removing punctuation.

```
...
... # Tokenize words
... tokenizer = Tokenizer(inputCol="clean_text", outputCol="words")
... tokenized_df = tokenizer.transform(clean_df)
...
... tokenized_df.select("words").show(5, False)
```

words
[the, project, gutenberg, ebook, of, the, project, gutenberg, gutenberg]
[encyclopedia, , vol, , , by, project, gutenberg]
[]
[this, ebook, is, for, the, use, of, anyone, anywhere, at, no, cost, and, with]
[almost, no, restrictions, whatsoever, , , you, may, copy, it, , give, it, away, or]
only showing top 5 rows

Tokenizing the cleaned text into individual words using Spark Tokenizer.

```
...
... # Remove stopwords
... remover = StopWordsRemover(inputCol="words", outputCol="filtered_words")
... filtered_df = remover.transform(tokenized_df)
...
... filtered_df.select("filtered_words").show(5, False)
...
```

```
+-----+
|filtered_words|
+-----+
|[project, gutenberg, ebook, project, gutenberg, gutenberg]|
|[encyclopedia, , vol, , , , project, gutenberg]|
|[]|
|[ebook, use, anyone, anywhere, cost]|
|[almost, restrictions, whatsoever, , , may, copy, , give, away]|
+-----+
only showing top 5 rows
```

Removing common stopwords from the tokenized text to prepare data for TF-IDF computation.

```
... # TF
... cv = CountVectorizer(inputCol="filtered_words", outputCol="tf_features")
... cv_model = cv.fit(filtered_df)
... tf_df = cv_model.transform(filtered_df)
...
... tf_df.select("tf_features").show(5, False)
```

```
+-----+
|tf_features|
+-----+
|(210917,[11,13,517],[3.0,2.0,1.0])|
|(210917,[0,11,13,4555,20603],[4.0,1.0,1.0,1.0,1.0])|
|(210917,[0],[1.0])|
|(210917,[110,517,676,852,1425],[1.0,1.0,1.0,1.0,1.0])|
|(210917,[0,15,41,80,159,363,1970,2781],[3.0,1.0,1.0,1.0,1.0,1.0,1.0,1.0])|
+-----+
only showing top 5 rows
26/02/08 00:47:02 WARN DAGScheduler: Broadcasting large task binary with size 3.3 MiB
```

Generating term frequency feature vectors for each book using the trained CountVectorizer model.

Term Frequency (TF) values were computed using the CountVectorizer transformation, which converts the filtered words of each book into a numerical feature vector representing the frequency of each term within the document.

```
... # IDF
... idf = IDF(inputCol="tf_features", outputCol="tfidf_features")
... idf_model = idf.fit(tf_df)
... tfidf_df = idf_model.transform(tf_df)
...
... tfidf_df.select("tfidf_features").show(5, False)
...
```

```

+-----+
|tfidf_features|
+-----+
|(210917,[11,13,517],[14.031615435522635,9.475626921014051,6.827788748744692])|
|(210917,[0,11,13,4555,20603],[1.3278235090076965,4.677205145174212,4.7378134605070255,8.994771533533466,11.18808985590262])|
|(210917,[0],[0.2819558772519241])|
|(210917,[110,517,676,852,1425],[5.613005028314541,6.827788748744692,7.0637891366811,7.271166595656634,7.696378466699633])|
|(210917,[0,15,41,80,159,363,1970,2781],[0.8458676317557723,4.775868362300808,5.150135197913108,5.463587848441875,5.840564701450054,6.6522885519342045,8.000577970327878,0.357977289524651])|
+-----+
only showing top 5 rows
26/02/08 00:47:02 WARN DAGScheduler: Broadcasting large task binary with size 3.3 MiB

```

Generating TF-IDF feature vectors for each book using the trained IDF model.

```

...
... # Normalize
... normalizer = Normalizer(inputCol="tfidf_features", outputCol="norm_features")
... norm_df = normalizer.transform(tfidf_df)
...
... norm_df.select("norm_features").show(5, False)
...

```

```

+-----+
|norm_features|
+-----+
|(210917,[11,13,517],[0.7685903000724685,0.5190333023387794,0.37399633517370146])|
|(210917,[0,11,13,4555,20603],[0.07109214852090885,0.2948267704900404,0.29864720455692156,0.566983777751028,0.7052391968677033])|
|(210917,[0],[1.0])|
|(210917,[110,517,676,852,1425],[0.36222303658301513,0.4406164543309324,0.45584622460939667,0.46922887659885043,0.4966689972371513])|
|(210917,[0,15,41,80,159,363,1970,2781],[0.84948970581313439,0.2794247130202082,0.30132217651305,0.3196615465026737,0.3417175666009214,0.3892096009584172,0.4680948120271521,0.4890053972071862])|
+-----+
only showing top 5 rows
26/02/08 00:47:02 WARN DAGScheduler: Broadcasting large task binary with size 3.3 MiB
26/02/08 00:47:02 WARN DAGScheduler: Broadcasting large task binary with size 3.3 MiB

```

Normalizing TF-IDF vectors of each book to prepare them for cosine similarity computation.


```

only showing top 5 rows
>>> from pyspark.sql.functions import desc
...
... # One vector per book
... book_vectors = norm_df.groupBy("file_name") \
...     .agg(first("norm_features").alias("norm_features"))
...
... # target vector
... target_vector = book_vectors \
...     .filter(col("file_name").contains("10.txt")) \
...     .select("norm_features") \
...     .head()[0]
...
... # cosine similarity
... cosine_udf = udf(lambda v: float(v.dot(target_vector)), DoubleType())
...
... similarity_df = book_vectors.withColumn(
...     "similarity",
...     cosine_udf(col("norm_features"))
... )
...
... # remove same book
... similarity_df \
...     .filter(~col("file_name").contains("10.txt")) \
...     .orderBy(desc("similarity")) \
...     .show(5, False)

```

```

+-----+-----+-----+
|file_name|similarity|norm_features|
+-----+-----+-----+
|file:///Users/sushantkparasharjha/D184MB/30.txt|0.2383177046644488|0.2414858810743713,0.28488332251424037,0.3618966396317069,0.3860073289937821,0.3877746627657421,0.421228857138091,0.4427130915867911|0.46486099898945627|
|file:///Users/sushantkparasharjha/D184MB/278.txt|0.210916,0.12,0.516,0.927,1292,1428|0.2882674807295836,0.2919927884073165,0.4207985866438226,0.4573702958346869,0.44983333318986726,0.4747594786006875|0.5738277837849314|
|file:///Users/sushantkparasharjha/D184MB/280.txt|0.210916,0.12,0.516|0.7685983800724685,0.5190333023387794,0.37399633517378146|0.5383752419971934|
|file:///Users/sushantkparasharjha/D184MB/279.txt|0.210916,0.12,0.516,0.927,1292,8387|0.2780727381617513,0.2735723955996386,0.39425244159832196,0.42851784658392614,0.43944432878788165,0.5657258238498657|0.5376278835802355|
|file:///Users/sushantkparasharjha/D184MB/432.txt|0.210916,0.12,0.516,0.927,1292,18170|0.2677494960246772,0.2712198564526624,0.39886098288168184,0.42483883416342116,0.43566410926194815,0.5759285283147237|0.5338029277566928|

```

Computing cosine similarity between the selected book and all other books using normalized TF-IDF feature vectors.

Identifying the top-5 books most similar to the selected book based on cosine similarity scores.

Each book was represented as a normalized TF-IDF vector capturing the importance of terms within the document. The cosine similarity between the selected book and all other books was computed using vector dot products. The books were then ranked according to similarity scores, and the top-5 most similar books were identified, providing a measure of textual similarity across the dataset.

Explain TF and IDF. Why is TF-IDF useful?

-> Term Frequency (TF) measures how often a word appears in a document, indicating the importance of that word within the specific document. Inverse Document Frequency (IDF) measures how rare a word is across the entire collection of documents; words that appear in many documents receive lower IDF values, while rare words receive higher values. TF-IDF

combines these two measures by multiplying TF and IDF, giving higher weights to words that are frequent in a particular document but rare across the corpus. This helps highlight meaningful keywords while reducing the influence of very common words.

How is cosine similarity calculated? Why is it appropriate for comparing documents represented as TF-IDF vectors?

-> Cosine similarity is calculated as the dot product of two document vectors divided by the product of their vector magnitudes. It measures the cosine of the angle between the vectors rather than their absolute length. When documents are represented as TF-IDF vectors, cosine similarity effectively measures how similar their word distributions are, regardless of document size. This makes it suitable for comparing text documents because longer documents do not automatically receive higher similarity scores.

Discuss scalability challenges when calculating pairwise cosine similarity for many documents. How could Spark help address these challenges?

-> Calculating pairwise cosine similarity for a large number of documents requires comparing every document with every other document, which results in very high computational and memory costs as the dataset grows. This process becomes difficult to handle on a single machine. Apache Spark addresses these challenges by distributing both data storage and computation across multiple nodes in a cluster, allowing similarity calculations to be processed in parallel. Spark's distributed DataFrame and ML libraries enable efficient large-scale vector processing, significantly reducing execution time for large document collections.

Question 12 - Author Influence Network

```
>>> from pyspark.sql.functions import *
...
... # Extract author + release date
... author_df = books_df.filter(
...     lower(col("text")).startswith("author:") |
...     lower(col("text")).startswith("release date:")
... ).groupBy("file_name").agg(
...     max(when(lower(col("text")).startswith("author:"), col("text"))).alias("author"),
...     max(when(lower(col("text")).startswith("release date:"), col("text"))).alias("release_date")
... )
...
... author_df.show(5, False)
```

```
file_name      author      release_date
file:///Users/sushantakparasharjha/D184MB/10.txt NULL      Release Date: March 2, 2011 [EBook #10]
file:///Users/sushantakparasharjha/D184MB/101.txt Author: Bruce Sterling      Release Date: January, 1994
file:///Users/sushantakparasharjha/D184MB/102.txt Author: Mark Twain (Samuel Clemens)      Release Date: January, 1994
file:///Users/sushantakparasharjha/D184MB/103.txt Author: Jules Verne      Release Date: May 15, 2008 [EBook #103]
file:///Users/sushantakparasharjha/D184MB/104.txt Author: Franklin Delano Roosevelt      Release Date: May 14, 2008 [EBook #104]
```

only showing top 5 rows

Extracting author and release date metadata from each book for constructing the influence network.

```

...
... # Extract year
... author_year_df = author_df \
...     .withColumn("year_str", regexp_extract(col("release_date"), "(\\d{4})", 1)) \
...     .filter(col("year_str") != "") \
...     .withColumn("year", col("year_str").cast("int"))
...
... author_year_df.show(5, False)
...

```

file_name	author	release_date	year_str	year
file:///Users/sushantakparasharjha/D184MB/10.txt	NULL	Release Date: March 2, 2011 [EBook #10]	2011	2011
file:///Users/sushantakparasharjha/D184MB/101.txt	Author: Bruce Sterling	Release Date: January, 1994	1994	1994
file:///Users/sushantakparasharjha/D184MB/102.txt	Author: Mark Twain (Samuel Clemens)	Release Date: January, 1994	1994	1994
file:///Users/sushantakparasharjha/D184MB/103.txt	Author: Jules Verne	Release Date: May 15, 2008 [EBook #103]	2008	2008
file:///Users/sushantakparasharjha/D184MB/104.txt	Author: Franklin Delano Roosevelt	Release Date: May 14, 2008 [EBook #104]	2008	2008

only showing top 5 rows

Extracting the release year from the release date field for influence analysis.

```

... # Influence edges
... X = 5
...
... influence_edges = author_year_df.alias("a").join(
...     author_year_df.alias("b"),
...     (col("b.year") > col("a.year")) &
...     ((col("b.year") - col("a.year")) <= X) &
...     (col("a.author") != col("b.author")))
... ).select(
...     col("a.author").alias("author1"),
...     col("b.author").alias("author2")
... )
...
... influence_edges.show(10, False)
...

```

author1	author2
Author: Bruce Sterling	Author: Project Gutenberg
Author: Bruce Sterling	Author: Robert Service
Author: Bruce Sterling	Author: Henry James
Author: Bruce Sterling	Author: Unknown
Author: Bruce Sterling	Author: Andrew Barton 'Banjo' Paterson
Author: Bruce Sterling	Author: Lao-Tze
Author: Bruce Sterling	Author: Joseph Conrad
Author: Bruce Sterling	Author: Joseph Conrad
Author: Bruce Sterling	Author: G. K. Chesterton
Author: Bruce Sterling	Author: Theodore Dreiser

only showing top 10 rows

Construction of directed author influence edges based on publication year difference within a 5-year window.

```

...
... # Out-degree
... out_degree = influence_edges.groupBy("author1") \
...     .count() \
...     .withColumnRenamed("count", "out_degree")
...
... out_degree.show(10, False)
...
...
... # In-degree
... in_degree = influence_edges.groupBy("author2") \
...     .count() \
...     .withColumnRenamed("count", "in_degree")
...
... in_degree.show(10, False)
...

```


only showing top 10 rows

author1	out_degree
Author: Horatio Alger Jr.	177
Author: Andrew Barton 'Banjo' Paterson	119
Author: Joanot Martorell and Marti Johan d'Galba	53
Author: Virginia Woolf	177
Author: Henry Lawson	13
Author: John Bunyan	13
Author: Michael Hart	53
Author: Jules Verne	255
Author: Various	325
Author: Giuseppe Salza	53

only showing top 10 rows

author2	in_degree
Author: Horatio Alger Jr.	12
Author: Andrew Barton 'Banjo' Paterson	178
Author: Joanot Martorell and Marti Johan d'Galba	56
Author: Henry Lawson	66
Author: Virginia Woolf	12
Author: John Bunyan	66
Author: Michael Hart	56
Author: H. B. Irving	116
Author: Jules Verne	130
Author: Giuseppe Salza	56

only showing top 10 rows

Computation of in-degree and out-degree for each author in the influence network.

```
...  
... # Top 5 most influential (OUT)  
... out_degree.orderBy(desc("out_degree")).show(5, False)  
...  
...  
... # Top 5 most influenced (IN)  
... in_degree.orderBy(desc("in_degree")).show(5, False)  
...
```

only showing top 10 rows

author1	out_degree
Author: Thomas Hardy	828
Author: Robert Louis Stevenson	817
Author: Edgar Rice Burroughs	720
Author: Henry James	613
Author: Frances Hodgson Burnett	551

only showing top 5 rows

author2	in_degree
Author: Robert Louis Stevenson	1911
Author: Edgar Rice Burroughs	941
Author: Arthur Conan Doyle	615
Author: L. Frank Baum	552
Author: Richard Harding Davis	512

only showing top 5 rows

Top 5 authors with the highest out-degree and in-degree showing the most influential authors and the most influenced authors based on the directed author influence network constructed using a 5-year publication window.

How was the influence network represented in Spark? What are the advantages and disadvantages of this representation?

-> The influence network was represented as a Spark DataFrame containing edges (author1, author2), where each row indicates that author1 potentially influenced author2 based on the release year condition.

Advantages

- DataFrames provide optimized query execution through Spark's Catalyst optimizer.
- Built-in aggregation functions allow easy computation of in-degree and out-degree.
- Scales efficiently for large datasets due to distributed processing.

Disadvantages

- DataFrame joins required for building the network can be computationally expensive for very large datasets.
- This representation does not directly support advanced graph algorithms unless converted to a graph framework such as GraphFrames.

How does the choice of the time window (X) affect the structure of the influence network?

-> The time window **X** determines how many connections are created between authors.

A small value of X results in fewer edges, producing a sparse network that captures only closely related publication periods.

A larger value of X creates more edges, producing a denser network where many authors appear connected, which may introduce less meaningful influence relationships.

What are the limitations of this simplified definition of influence?

-> This approach assumes influence based solely on publication years, which does not capture actual intellectual influence between authors. True influence depends on citations, literary style, historical context, and collaboration, none of which are considered here. Therefore, the network provides only an approximate representation of potential influence.

How well would this approach scale to a much larger dataset? What optimizations could be considered?

-> The approach can scale to large datasets because Spark distributes computation across multiple nodes. However, the self-join used to generate edges may become expensive as the number of authors increases.

Possible optimizations include:

- Partitioning the dataset by publication year to reduce join cost.
- Filtering invalid or missing metadata before network construction.
- Using broadcast joins when one dataset is small.

- Employing graph-processing frameworks such as GraphFrames for efficient large-scale network analysis.